drop per unit length of wire. So the last segment has the largest potential drop per unit length.

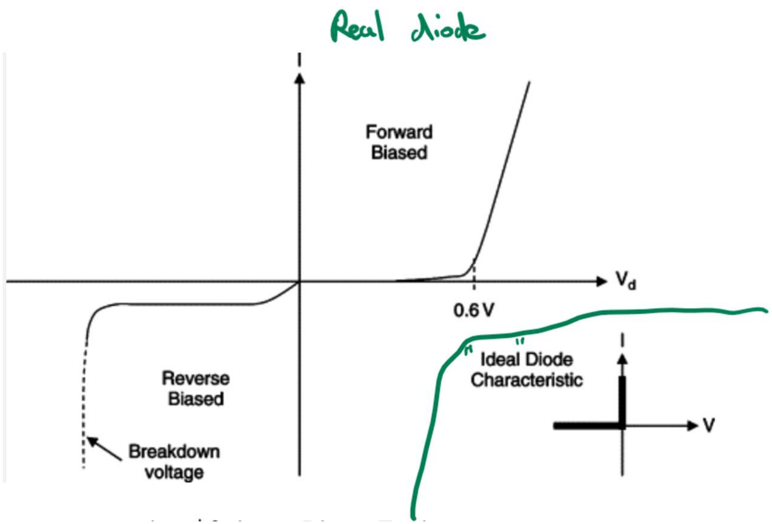
21 Ans: A

Since the thermistor's resistance decreases as its temperature increases, in accordance with the potential divider principle, the potential difference (p.d.) across it decreases with increasing temperature.

The thermistor and the fixed resistor are in series thus their potential differences add up to 10 V. Therefore the p.d. across the fixed resistor increases as the p.d. across the thermistor decreases due to increase in temperature.

When the p.d. across the fixed resistance exceeds the threshold p.d. of the diode (normally around 0.3 – 0.7 V) the diode becomes forward biased and current starts to flow downwards through the lamp to light it up.

Option C is wrong as the lamp will light up immediately regardless of temperature. (It will get brighter as the temperature increases.)



22 Ans: C